where n is the number of moles and V is the volume of the container (dm^3 or L)

8.6.1 Equilibrium constants (K_p and K_c):

Let us consider a reversible reaction,

$$xA + yB \rightleftharpoons lC + mD$$

where, A and B are the reactants, C and D are the products and x,y,*l* and m are the stoichiometric coefficients of A, B, C and D, respectively.

Applying the law of mass action, the rate of the forward reaction,

$$r_f \alpha [A]^x [B]^y \text{ (or) } r_f = k_f [A]^x [B]^y$$

Similarly, the rate of the backward reaction,

$$r_b \alpha [C]^l [D]^m$$

(or)

$$r_b = k_b [C]^l [D]^m$$

where k_f and k_b are proportionality constants

At equilibrium,

Rate of forward reaction (r_f)

= Rate of backward reaction (r_b)

$$k_f [A]^x [B]^y = k_b [C]^l [D]^m$$

or $$\frac{k_f}{k_b} = \frac{[C]^l [D]^m}{[A]^x [B]^y} = K_c$$

where, K_c is the equilibrium constant in terms of concentration (active mass).

At a given temperature, the ratio of the product of active masses of reaction products raised to the respective stoichiometric coefficients in the balanced chemical equation to that of the reactants is a constant, known as equilibrium constant. Later when we study chemical kinetics we will learn that this is only approximately true.

If the reactants and products of the above reaction are in gas phase, then the equilibrium constant can be written in terms of partial pressures as indicated below,

$$K_p = \frac{p_C^l \times p_D^m}{p_A^x \times p_B^y}$$

Where, p_A, p_B, p_C,and p_D are the partial pressures of the gas A, B, C and D, respectively.

8.6.2 Relation between K_p and K_c

Let us consider the general reaction in which all reactants and products are ideal gases.

$$xA + yB \rightleftharpoons lC + mD$$

The equilibrium constant, K_c is

$$K_c = \frac{[C]^l [D]^m}{[A]^x [B]^y} \qquad (1)$$

and K_p is,

$$K_p = \frac{p_C^l \times p_D^m}{p_A^x \times p_B^y} \qquad (2)$$

The ideal gas equation is

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